

## Indices Worked Examples

Law of Indices referred to here are explained in the accompanying PowerPoint Presentation.

### Example 1:

Simplify the following:

$$\frac{(y^3 \cdot y^{\frac{1}{5}})}{y^{\frac{1}{2}}}$$

Firstly multiply  $(y^3 \cdot y^{\frac{1}{5}})$  by applying Law 1 of Indices- add the indices:

$$y^{3+\frac{1}{5}} = y^{\frac{16}{5}}$$

Then we divide  $y^{\frac{16}{5}}$  by  $y^{\frac{1}{2}}$  by applying Law 2 of Indices - subtract the indices:

$$y^{\frac{16}{5} - \frac{1}{2}} = y^{\frac{27}{10}}$$

### Example 2:

Simplify the following:

$$\frac{\sqrt{x^4} \cdot \sqrt{x^2}}{x^{-2}}$$

Firstly we deal with  $(\sqrt{x^4} \cdot \sqrt{x^2})$  by applying Law 6 of Indices:

$$x^{\frac{4}{2}} \cdot x^{\frac{2}{2}} \text{ This can also be written as } x^2 \cdot x^1 = x^{2+1} = x^3$$

We then divide  $x^3$  by  $x^{-2}$  by subtracting the indices  $x^{3-(-2)} = x^{3+2} = x^5$

### Example 3:

Simplify the following expressing the answer with positive indices only:

$$\frac{(ab^2)^3}{\left(a^{\frac{1}{2}} b^{\frac{1}{4}}\right)^4} =$$

Applying Law 3 of Indices:

$$(a^1 b^2)^3 = a^{1 \times 3} b^{2 \times 3} = a^3 b^6 .$$

$$\left(a^{\frac{1}{2}} b^{\frac{1}{4}}\right)^4 =$$

for  $a$ : multiplying it's index  $\frac{1}{2}$  by 4  $\rightarrow a^2$ ;

for  $b$ : multiplying it's index  $\frac{1}{4}$  by 4  $\rightarrow b^1$

$$\text{giving us } \left(a^{\frac{1}{2}} b^{\frac{1}{4}}\right)^4 = a^2 b^1.$$

$$\text{Now, for } \frac{a^3 b^6}{a^2 b^1} =$$

Applying Law 2 of Indices:

$$a^{3-2} = a^1 = a$$

$$b^{6-1} = b^5$$

$$\text{Answer: } \frac{(ab^2)^3}{\left(a^{\frac{1}{2}} b^{\frac{1}{4}}\right)^4} = ab^5$$